Vinayak Sharma

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EDUCATION

[PhD] Doctor of Philosophy in Computer Science — Present
Arizona State University, Tempe, AZ — Current GPA: 4.0/4.0

[M.S] Master of Science in Computer Science — May 2024
Arizona State University, Tempe, AZ — GPA: 4.0/4.0

[B.Tech] Bachelors of Technologyin Computer Science and Engineering — July 2021
SRM Institute of Science & Technology, Chennai, TN, India — GPA: 8.98/10

RESEARCH EXPERIENCE

MPS-Lab | *Graduate Research Assistant* — Jan 2023 - Present

- Led a joint research project between Arizona State University (ASU) and the National Institute of Technology (NIT) - Rourkela, on Quantum Machine Learning.
- Taught sessions on Quantum Machine Learning at NIT Rourkela, VLSI Conference 2025, ES-Week 2024.

PUBLICATIONS

- **Real-Time Trust Verification for Safe Agentic Actions using** , AAAI 2026 Workshop on Trust and Control in Agentic AI (TrustAgent),
- **QuIRK: Quantum-Inspired Re-uploading KAN**, ArXiV Preprint 2025,
- **QPMeL - Quantum-Aware Classically-Trained Embeddings via Projective Metric Learning**, ArXiV Preprint 2025,
- **LUCI: Lightweight UI Command Interface**, Proceedings of Languages, Compilers, Tools and Theory of Embedded Systems (LCTES), 2025

INDUSTRIAL EXPERIENCE

Anton Intelligence, Tempe ,AZ | *Artificial Intelligence/Machine Learning Intern, PhD* — May 2026 - Present

- Designing and developing software and **AI/ML systems** to advance the company's AI-powered software platform.
- Conducting **applied research** and proof-of-concept development to evaluate new techniques and technologies for integration into production software.
- Providing **research-informed guidance** to the broader engineering team regarding system architecture, model selection, and implementation approaches.

Product Engineer: Machine Learning | *Myelin Foundry,Bengaluru,India* — Aug 2021 - Jun 2022

- Overhauled inference pipeline for a multi-model orchestration system comprising of **face detection, recognition & Driver management** models resulting in **a 70% increase in the number of running concurrent models**.
- Optimized computer vision system utilizing Tensorflow with TensorRT acceleration on Nvidia Jetson platform, resulting in a **2x increase in inference speed** and **50% increase in defect detection rate** in real-time steel manufacturing.
- Reduced footprint of a *low-light image enhancement* model by 50% by adapting the **ZeroDCE++ architecture to run faster on a DSP** enabling deployment on embedded devices.
- Created a new architecture and training pipeline for a **joint denoise & super-resolution (SR) model** *reducing compression artifacts by 60% for Super Resolution* outputs on encoded video streams.
- Awarded the **Myelin Impact Award** as the *most impactful employee* for the company in the year 2021.

Samsung Research Institute, Bengaluru | *PRISM ML Research* — Nov 2019 - Aug 2020

- Developed an **LSTM based model which used Bezier Curves** to perform *online handwriting recognition* **50% faster** without any loss in accuracy.
- Collaborated with a team from Samsung to collect **over 1300 samples of Bengali handwriting** for a custom dataset for the Bengali Language.
- Awarded the title **'Excellent'** given only to the top 5% of teams

TEACHING EXPERIENCE

Graduate Teaching Assistant, Quantum Computation | *Arizona State University* — Spring 2025,Fall 2023

- Created and graded assignments, quizzes, and exams for the course.
- Taught the section on Quantum Machine learning with a focus of variational, Kernel and Data Relploading models.

SKILLS

- **Machine Learning Frameworks:** PyTorch, TensorFlow, Keras, JAX, Scikit-Learn, OpenCV, TF-Lite,
- **Quantum Computing Frameworks:** Qiskit, Cirq, PennyLane